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Sand Dam Vegetation Impact Assessment, Machakos, Kenya

Site: sand dam at -1.5435°, 37.3326° (Athi-Kapiti plains, Machakos County) **Constructed:** October 2018 **Analysis period:** 2016–2025 **Season:** long dry season only (June to September) **Method:** within-AOI buffered-corridor distance-decay analysis from Sentinel-2 satellite time series; pre-construction vs post-construction comparison **Author / Date:**

Executive Summary

This assessment examines whether a sand dam built in October 2018 at -1.5435°, 37.3326° in Machakos County, Kenya, has changed the plant cover at the dam site. The evidence is drawn from satellite images of the area between 2016 and 2025, restricted to the long dry months (June to September) when the dam’s water-storage role is most visible.

Key Findings

The patterns in the satellite record show three lines of evidence that the dam is having an effect on the surrounding land. Each finding is described in plain language here, with the underlying analysis in §3.

- **The riverbed itself is the most visibly changed zone in the area.** Across the full length of the river through the dam site, the strip of land 0 to 50 metres from the channel lost about 28 percentage points of its dry-season green cover after the dam was built. This pattern is concentrated on the river itself, with the largest losses on the downstream arm of the channel where the dam wall does not stop sand. The pattern is consistent with what a working sand dam produces from above (sand accumulating in the channel reading as bare ground), but at this site the largest channel reduction is on the downstream side, which can also be explained by the channel-side vegetation losing the most in absolute terms during the drought.
- **The strip of land alongside the upstream arm of the river is holding its plant cover better than anywhere else nearby.** On the upstream side of the dam wall (the arm of the river that water flows down towards the dam), the band of land 100 to 250 metres from the channel lost only 14 percentage points of plant cover, the smallest reduction in any zone of the upstream side. The most likely reason is that the dam holds water in the sand of the upstream channel, supporting plants along the riverbanks during the dry months.
- **Land far from the river declined the most, in line with the drought.** Areas 500 to 1,000 metres from the river on the upstream side, where the dam's water cannot reach, show plant-cover losses that match what would be expected from the regional drought. The fact that closer-to-river land on the upstream side declined less than this is what makes the dam's effect visible there.

These findings describe how plant cover changed at the dam site after construction, *during a period dominated by the most severe drought the region has seen in over 40 years*. Annual rainfall at the site fell from 1,011 mm in pre-drought years to 717 mm during the drought, a reduction of 29 per cent. The patterns above describe how the dam's location appears to have shaped the local response to that drought; they do not, on their own, say how much vegetation in absolute terms the dam preserved.

What This Assessment Does Not Show

The assessment does not estimate how much vegetation the dam has preserved compared to a scenario in which it had not been built. To produce that figure, an analysis of one or more matched comparison areas (without dams) would be required, following the procedure described in §5. The current analysis instead documents the spatial pattern of change at the dam site itself, which shows the *fingerprint* of the dam's hydrology even if it does not measure the size of the effect.

Numbers Underlying the Findings

TABLE 1. Change in how often each zone of land was reliably green during the long dry season, after the dam was built (2019–2025) compared to before (2016–2018). Negative values indicate a reduction in green cover; the most negative values are the largest losses.

Distance from river	Combined upstream + downstream	Upstream of dam wall (WSW)	Downstream of dam wall (NNE)
0–50 m (the riverbed itself)	-27.9 pp	-20.2 pp	-34.9 pp
50–100 m	-20.8 pp	-17.3 pp	-24.8 pp
100–250 m	-16.7 pp	-14.3 pp	-20.4 pp
250–500 m	-13.2 pp	-19.0 pp	-12.2 pp
500–1,000 m	-10.7 pp	-22.8 pp	-13.6 pp

'pp' means percentage points. A value of -27.9 pp means the share of the area that was reliably green during the long dry season fell by 27.9 percentage points after the dam was built (for example, from 71 per cent of the area down to 43 per cent).

The contrast between the upstream and downstream columns is the visible asymmetry described in §3.4. Direction along the river was confirmed by elevation lookup against the SRTM 30 m digital elevation model: the WSW arm of the river rises 9 metres from the dam to its far end

and is therefore the upstream side; the NNE arm is essentially flat (a 1 metre difference, within DEM noise) and is the downstream side.

1. Site and Approach

The sand dam is located at -1.5435° , 37.3326° , in the Athi-Kapiti plains of Machakos County, approximately 50 km southeast of Nairobi. The area lies within Kenya's semi-arid lands, with bimodal rainfall (long rains March to May, short rains October to December). The dam was constructed in October 2018 across a seasonal sandy streambed.

A sand dam is a small concrete wall built across a seasonal riverbed. Sand carried by floods accumulates upstream of the wall over successive rainy seasons. The accumulated sand stores water in its interstitial spaces, raising the local water table along the channel and providing dry-season water access for vegetation rooted within the riparian zone.²

The treatment area of interest (AOI) is a $1 \text{ km} \times 1 \text{ km}$ square centred on the dam. The river through the AOI was hand-traced from satellite imagery for the corridor analysis described in §3.3 and §3.4. Figure 1.1 shows the trace overlaid on the satellite picture (left panel) and the distance bands used in the analysis (right panel). Full technical detail of how the trace was produced is in §9.1.4.

FIGURE 1.1. Site and methodology. Left: hand-traced channel polyline (cyan) and dam location (white star) on the mean post-construction NDVI of the treatment AOI. Right: distance bands around the channel.

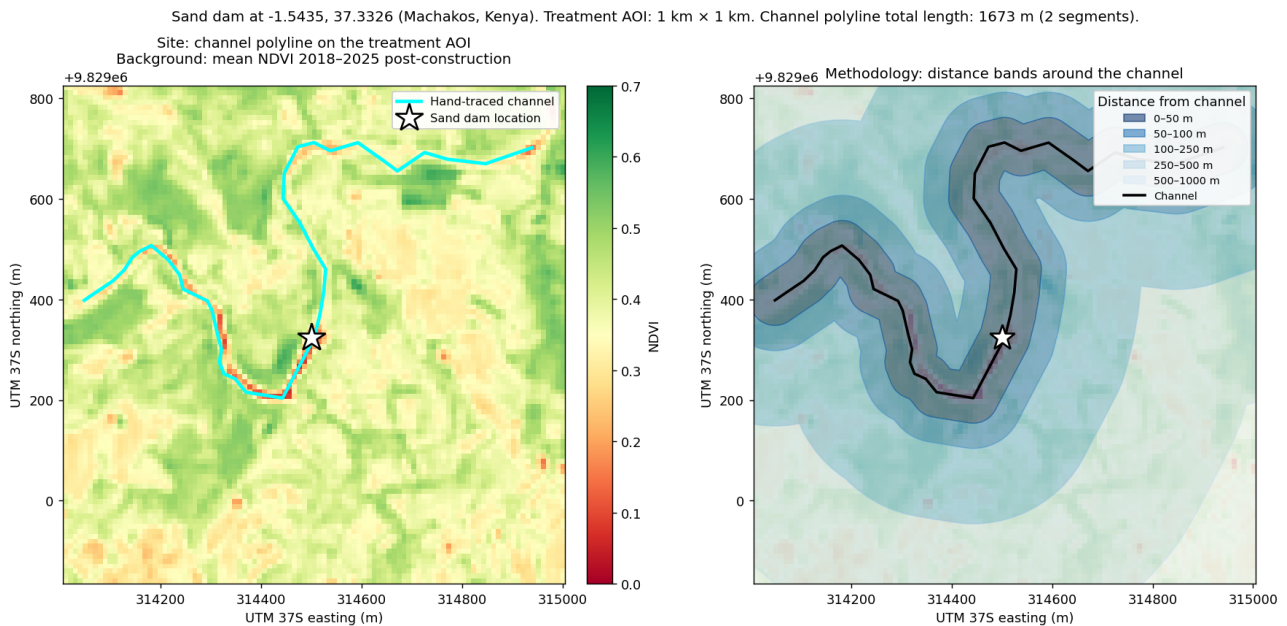


Figure 1: Site map and methodology rings

Source: Sentinel-2 L2A (Microsoft Planetary Computer); channel polyline by manual interpretation.

2. Climate Context

2.1 Regional Setting

Machakos County experiences a semi-arid climate with bimodal rainfall. The long rains fall between March and May, the short rains between October and December. Mean annual rainfall in the analysis period was 1,011 mm in pre-drought years (2016 to September 2020), with marked seasonal and inter-annual variation. The two dry seasons (June to September and January to

February) are the most stressful periods for vegetation and the period when sand-dam interventions are designed to deliver the most value.

2.2 The 2020–2023 Horn of Africa Drought

The 2020–2023 Horn of Africa drought was the most severe in the region for at least 40 years.¹ Five consecutive failed rainy seasons, from October 2020 through May 2023, were associated with an unusually persistent triple-dip La Niña event in the Pacific. The drought affected approximately 36 million people across Kenya, Ethiopia, and Somalia. The Kenya National Drought Management Authority classified Machakos County in ‘alert’ or ‘alarm’ phase through most of 2022.³

In this analysis, the drought window is defined as 1 October 2020 to 30 June 2023. Mean annual rainfall at the dam site during the drought was 717 mm, a reduction of 29 per cent from the pre-drought mean.

The drought is the dominant feature of the analysis period. Because of it, any straightforward ‘before-and-after’ comparison of plant cover at the dam site would mostly measure the drought rather than anything the dam was doing. To get around this, the analysis instead looks at how the change in plant cover is distributed *within* the dam site: if the dam is having an effect, the change should be concentrated near the river, not spread evenly across the whole area. Section §3.3 sets out this test in detail.

2.3 Annual Dry-season Vegetation Cover, 2016–2025

Figure 2.1 shows the median NDVI of the treatment AOI for each year of the analysis period, computed from long-dry-season scenes only (June to September). Pre-construction years (2016, 2017, 2018) appear moderately green. The 2019 and 2020 panels show comparable greenness. The 2021, 2022 and 2023 panels become progressively browner; 2022 and 2023 are visibly the driest years in the series. The 2024 panel shows a partial recovery following the strong El Niño rains of late 2023. The 2025 panel shows a return to drier conditions.

FIGURE 2.1. Annual median NDVI in the treatment AOI during the long dry season (June to September), 2016–2025.

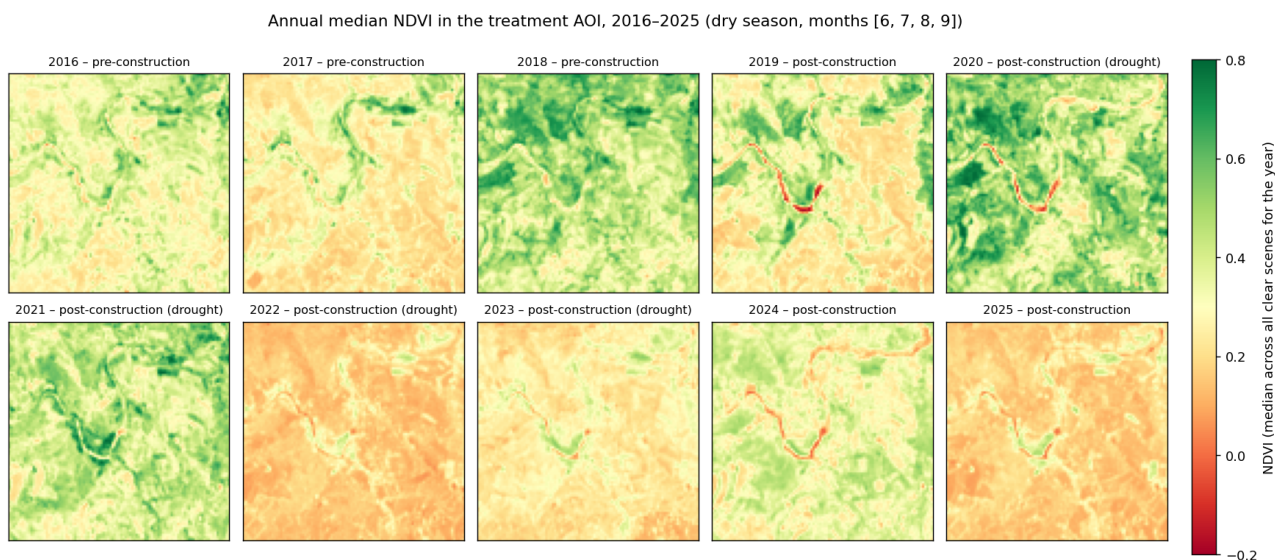


Figure 2: Annual median NDVI grid

Source: Sentinel-2 L2A (Microsoft Planetary Computer), IEP calculations.

Figure 2.1 establishes the climate context for the analysis: dry-season vegetation cover at the site has varied substantially across the period and the variation is dominated by the drought. Any straightforward before-after comparison will be heavily influenced by which years fall on each side of the construction date.

3. Results

3.1 Plant Cover and Rainfall Across the Period

Figure 3.1 shows two views of the dam site over the full ten years. The top panel shows how green the area was at the time each cloud-free satellite image was taken (the dots), with a smoothed line showing the average over each 90-day window. The bottom panel shows monthly rainfall.

FIGURE 3.1. Plant cover and rainfall at the dam site, 2016–2025. Dam construction date (October 2018) is marked with a dashed line; the 2020 to 2023 drought period is shaded pink.

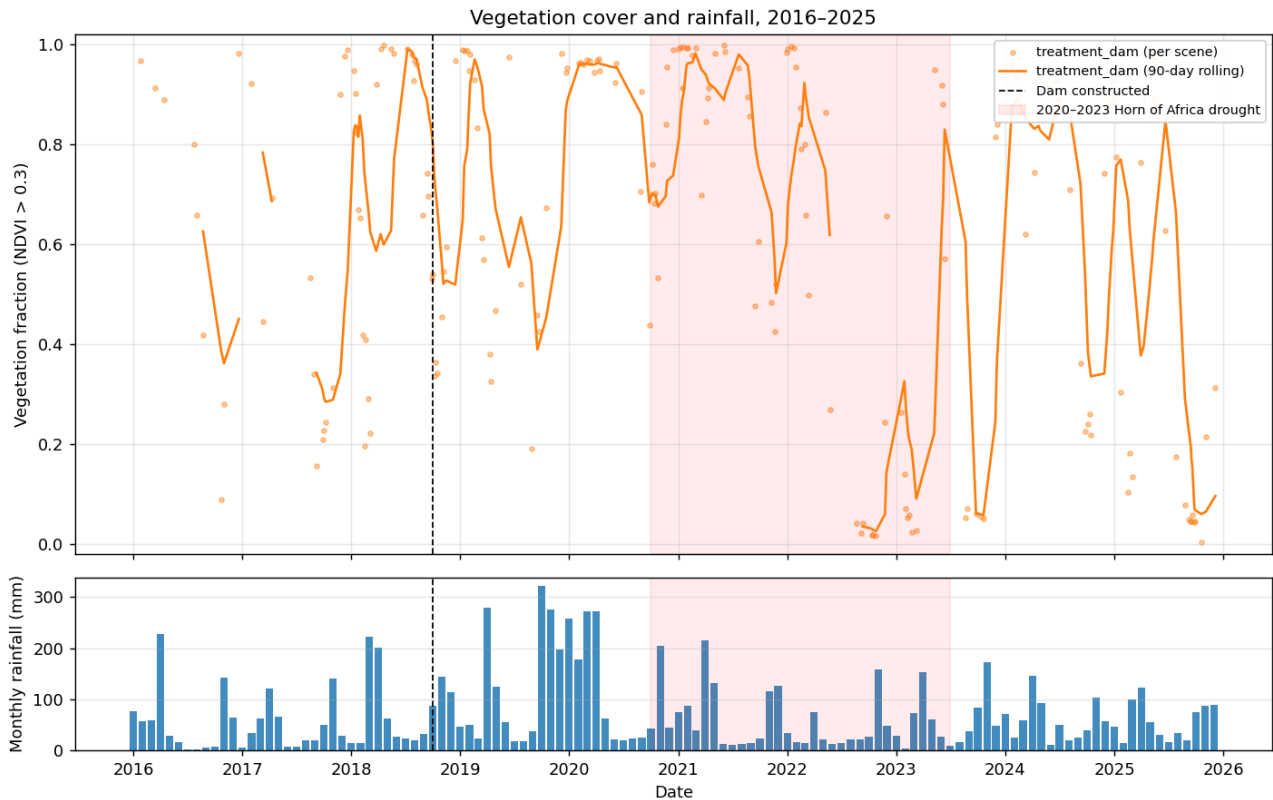


Figure 3: Time series with precipitation

Source: Sentinel-2 L2A (Microsoft Planetary Computer); ERA5 reanalysis via Open-Meteo Historical Weather API; IEP calculations.

Three things stand out in Figure 3.1.

First, the area shows a strong seasonal cycle: plant cover peaks during the rainy seasons and drops during the dry seasons, year after year. The cycle is the dominant feature of the chart at any timescale shorter than several years.

Second, for the first two years after the dam was built, the site continued to behave much as it had before construction. The dam has a slow effect, because sand needs time to accumulate behind the wall.

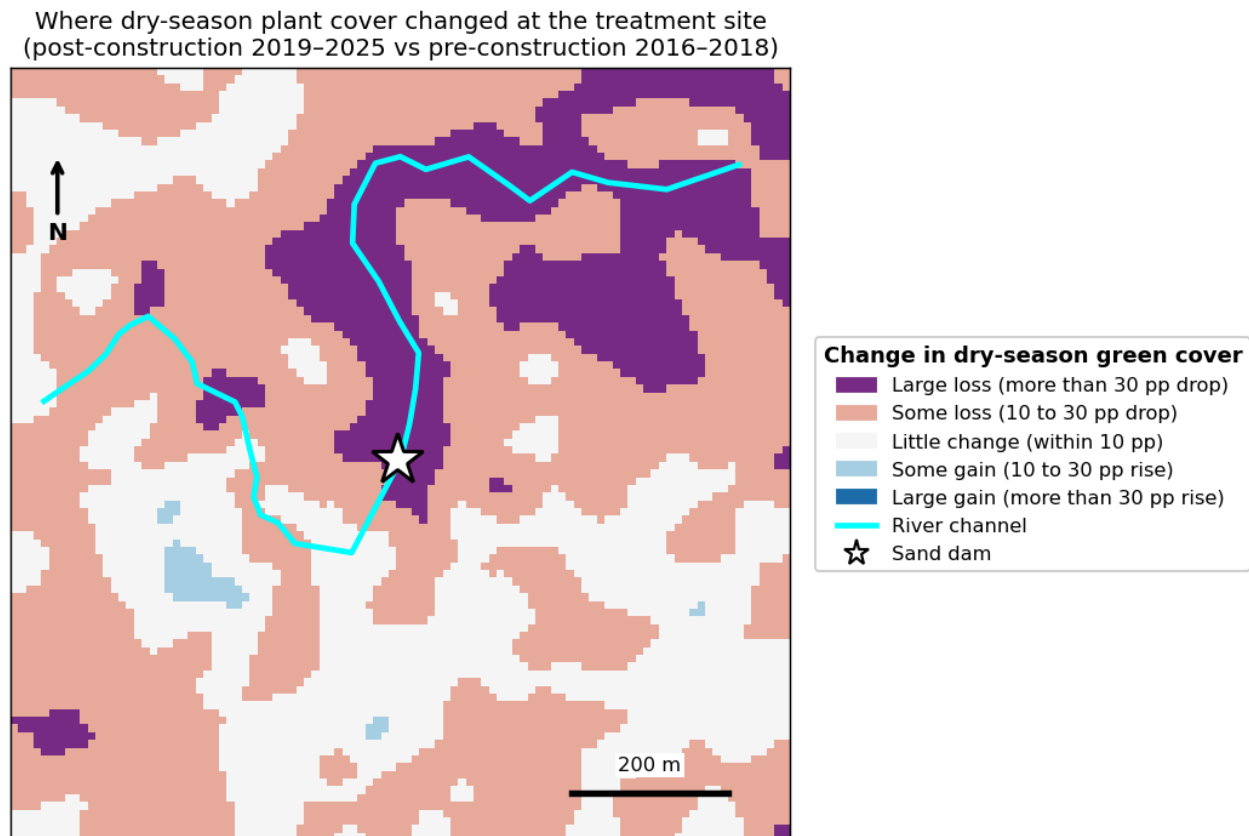
Third, from late 2020 onwards (when the drought period begins), rainfall drops below earlier years and plant cover follows. The deepest plant-cover troughs are in 2022 and again in 2025.

The chart makes clear why a simple before-and-after comparison of plant cover at the site would not be a fair test of whether the dam works. The drought arrived two years after the dam was built, so almost the entire ‘after dam’ period is also an ‘in drought’ period. Any comparison that ignores the drought will mistake the regional climate signal for an effect of the dam. The corridor analysis in §3.3 and §3.4 sidesteps this by looking at how the change is distributed within the dam site itself, not at the change as a whole.

3.2 Where the Change Has Happened on the Map

Figure 3.2 shows where, within the dam site, plant cover during the dry season has changed after the dam was built compared to before. Each square of land is shaded by category: dark purple for a large loss in dry-season green cover, light pink for some loss, light grey for little change, and shades of blue for gain. The cyan line traces the river through the area, and the white star marks the dam itself.

FIGURE 3.2. Map showing where dry-season plant cover changed at the dam site, after the dam was built (2019–2025) compared to before (2016–2018).



Each square of land is shaded by how much its dry-season green cover changed after the dam was built. 'pp' means percentage points (a 10 pp drop is roughly: from 60 per cent of the time green to 50 per cent). Light spatial smoothing has been applied to reduce single-pixel noise.

Figure 4: Per-pixel change in treatment AOI

Source: Sentinel-2 L2A (Microsoft Planetary Computer), IEP calculations.

Two features stand out.

First, the dark purple band, which marks the largest loss of dry-season green cover, sits directly along the river, with a particular concentration upstream of the dam (the part of the river above the white star). Light pink, which marks some loss, surrounds the river on both sides. Most of the area away from the river is light grey, meaning little change.

Second, a few patches of light blue (some gain) appear to the south and southwest of the dam, away from the river itself. These are small areas where dry-season green cover has actually increased after the dam was built.

The pattern of loss concentrated on the river is what would be expected from satellite imagery of a working sand dam. There are three explanations, all of which produce the same kind of pattern and which the satellite data alone cannot tell apart:

1. **Sand piling up behind the dam wall.** A working sand dam captures sand carried by

floods. The accumulating sand reads as bare ground from satellite, so the riverbed appears 'less green' even though the dam is functioning as designed. This is the simplest interpretation of the dark purple band on the river.

2. **Drought hitting the greenest land hardest.** Land closest to the river was the greenest land before the dam was built. When a drought reduces plant cover in absolute terms, areas with more plants to lose end up losing more. Some of the channel-aligned loss will reflect this, regardless of whether the dam is doing anything.
3. **More dry-season farming near the dam.** A working sand dam often enables farmers to grow crops in dry months that they could not have grown before. Freshly tilled fields read as bare ground from satellite, so successful dam-enabled farming would also produce loss along the river.

A site visit, with photographs of the streambed and conversations with farmers about cropping practices, would distinguish between these three explanations. From satellite imagery alone, it is not possible to tell which contributes most. §4 returns to this point.

3.3 How the Change Falls Off with Distance from the River

To put numbers on the pattern in Figure 3.2, the area was divided into concentric distance bands around the river: 0 to 50 metres, 50 to 100 metres, 100 to 250 metres, 250 to 500 metres, and 500 to 1,000 metres. For each band, the average loss of plant cover during the dry season was calculated. Figure 3.3 shows the result.

FIGURE 3.3. Loss of dry-season plant cover at the dam site, by distance from the river.

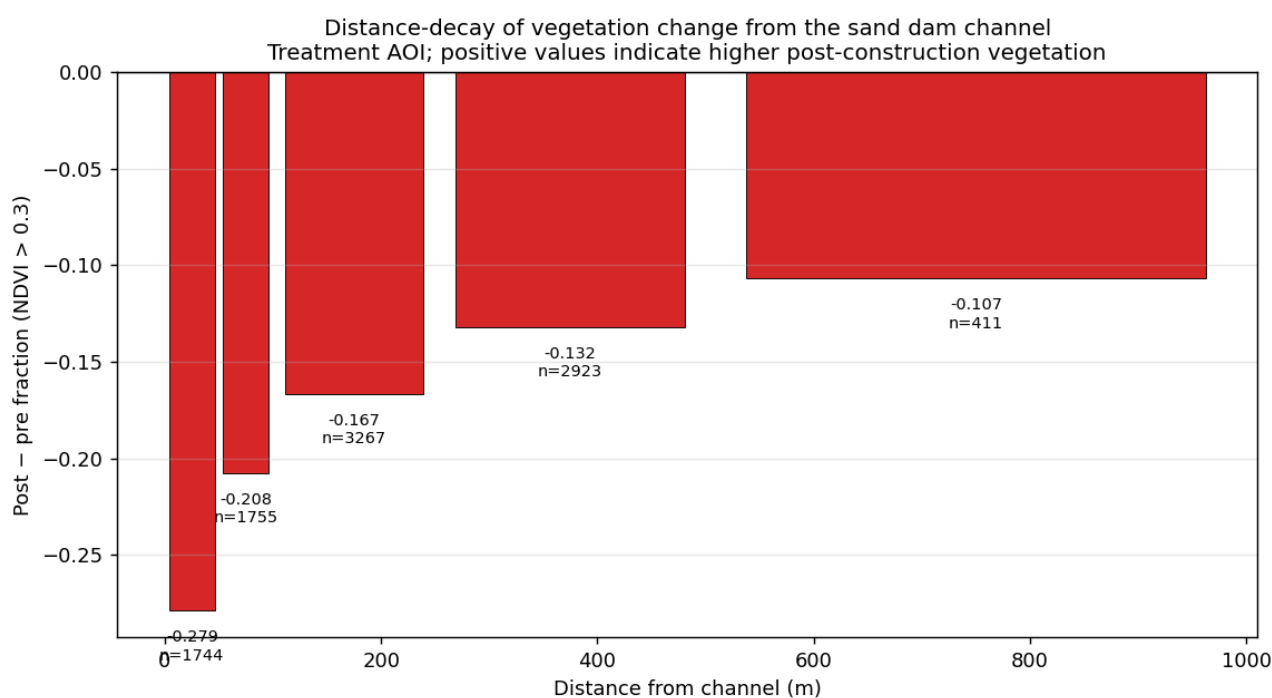


Figure 5: Distance-decay corridor chart

Source: Sentinel-2 L2A (Microsoft Planetary Computer); channel polylines by manual interpretation; IEP calculations.

The numbers behind the chart are in Table 2.

TABLE 2. Average dry-season plant-cover loss in each distance band.

Distance from river	Number of 10 m squares	Plant-cover share before dam	Plant-cover share after dam	Reduction
0–50 m (the riverbed itself)	1,744	71 per cent	43 per cent	-27.9 pp
50–100 m	1,755	66 per cent	45 per cent	-20.8 pp
100–250 m	3,267	60 per cent	44 per cent	-16.7 pp
250–500 m	2,923	57 per cent	44 per cent	-13.2 pp
500–1,000 m	411	58 per cent	47 per cent	-10.7 pp

The pattern is consistent and gradual: the closer to the river, the larger the loss. The riverbed itself lost 27.9 percentage points of its dry-season plant cover, while land 500 to 1,000 metres away lost 10.7 percentage points. There is no single point where the pattern jumps; the loss eases off smoothly as you move from the river outwards.

This is the signature of an effect that is concentrated on the river. A drought that hits the whole area uniformly would not produce a smooth gradient like this; nor would random noise in the satellite measurements. The pattern is therefore evidence that the river itself is where the change is happening. As §3.2 explains, the satellite imagery alone cannot say whether that change is sand piling up, more farming, or differential drought response – but it can show that the change is anchored to the river.

What the analysis does not show, on its own, is how much vegetation was preserved compared to a no-dam scenario. That figure would require a comparable area without a dam to be analysed in parallel, following the procedure described in §5.

3.4 The Land Above and Below the Dam Wall Behaves Differently

A sand dam is a wall built across a river. It affects the land on each side of the wall differently. On the **upstream** side (above the wall, where water flows down towards the wall), the dam captures sand carried by floods. The sand piles up behind the wall over years, and the water held in that sand keeps the surrounding land moist. On the **downstream** side (below the wall, where water flows away from the dam), there is no sand captured, but the dam slowly releases water through the dry months, helping support plants alongside the river.

For this site, the upstream and downstream directions were determined by checking elevations along each arm of the river against the SRTM 30 m digital elevation model. The arm running west-southwest from the dam rises 9 metres over its length and is therefore upstream of the wall (water comes from there). The arm running north-northeast is essentially flat (1 metre difference, within DEM noise) and is downstream of the wall.

To check whether the satellite record shows different patterns on each side, the corridor analysis in §3.3 was repeated separately for the two arms. The comparison is shown in Figure 3.4. This is *not* a comparison of a treated area against a non-treated control: both sides of the wall are part of the dam’s influence zone. It is a comparison of how the dam shows up differently on each side.

FIGURE 3.4. Loss of dry-season plant cover, broken down by which side of the dam wall the river segment is on. The upstream arm runs west-southwest from the dam; the downstream arm runs north-northeast.

Source: Sentinel-2 L2A (Microsoft Planetary Computer); channel polylines by manual interpretation; flow direction by SRTM 30 m DEM lookup; IEP calculations.

The two sides behave differently:

- **The upstream side shows a U-shape.** The upstream riverbed lost 20 percentage points of plant cover. The strip of land 100 to 250 metres from the river lost the least of any zone on the upstream side, only 14 percentage points. The land 500 to 1,000 metres from the river lost the most, around 23 percentage points. This pattern is consistent with the dam holding water in the upstream sand bed and supporting riparian vegetation along that arm: the strip of land closest to the river holds its plant cover better than land further away. The riverbed itself shows a moderate loss rather than the very large loss that would be expected

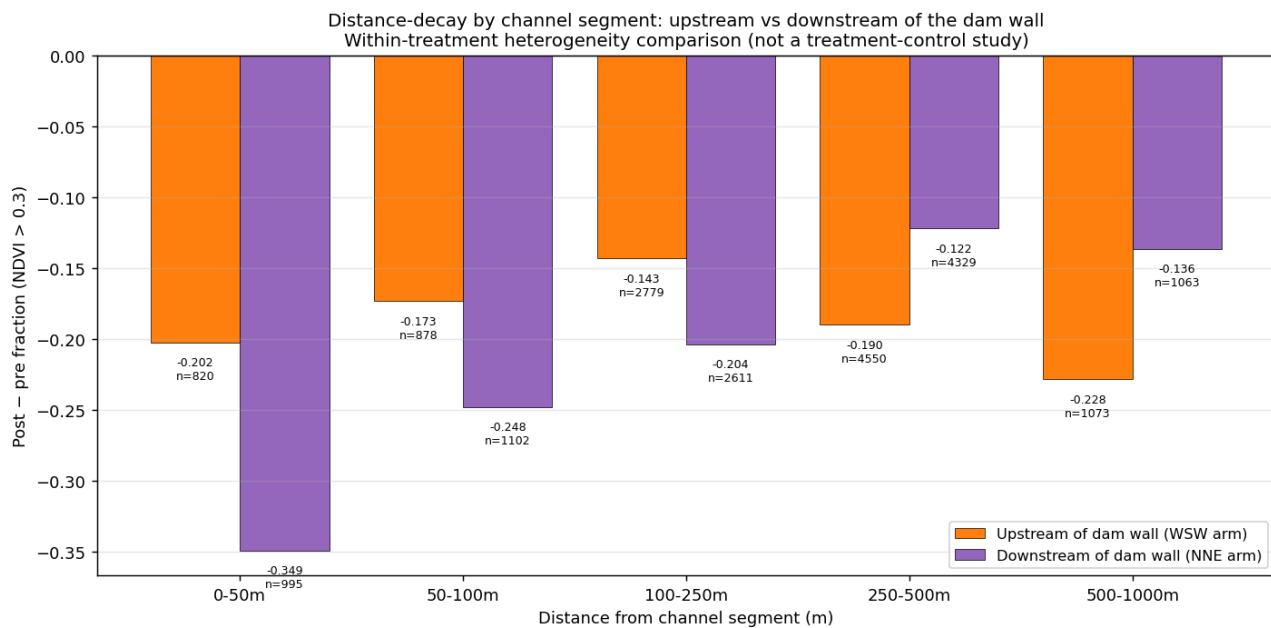


Figure 6: Corridor decay by segment

if sand were piling up at the surface; sand may still be accumulating below the surface, or sand fill may not yet be the dominant signal in the satellite record.

- **The downstream side shows a different pattern, a smooth gradient outwards from the river.** The downstream riverbed lost the most plant cover anywhere in the area, around 35 percentage points. From there, the loss steadily eases off as you move away from the river, down to about 14 percentage points at 500 to 1,000 metres. This is a striking pattern but it is harder to explain solely through the dam mechanism: the downstream side of the wall should not accumulate additional sand. The most likely explanations are that this arm of the river had the greenest pre-construction vegetation and lost the most in absolute terms during the drought, or that dry-season cropping enabled by the dam has expanded along the downstream channel where farmers can now access shallow water.

The two different signatures – one with a smallest-loss zone in the upstream riparian band, the other with a smooth gradient anchored to the downstream channel – are both consistent with the channel being the structuring feature of the change, but they suggest different mixes of mechanisms on each side. Distinguishing between sand accumulation, drought response, and dry-season cropping requires ground-truth observation, as set out in §4.

4. Limitations

4.1 No Counterfactual Magnitude Estimate

The patterns described in §3.3 and §3.4 are evidence that something is happening at the dam site that is anchored to the river. They do not, on their own, say *how much* vegetation the dam has preserved. To produce a statement such as ‘the dam has preserved X hectares of vegetation’, a comparable area without a dam needs to be analysed in parallel and the difference between the two attributed to the dam. This assessment did not have access to such a comparison area; what would be required to set one up is described in §5.

4.2 NDVI Bias in Semi-arid Landscapes

NDVI is known to be sensitive to soil brightness and moisture in addition to vegetation cover, and to saturate in dense vegetation.⁶ In semi-arid landscapes such as Machakos, the soil-brightness sensitivity is the more relevant limitation. Freshly tilled cropland and wet bare soil after rains can both shift NDVI in directions that complicate interpretation. The Soil-Adjusted Vegetation

Index (SAVI) and the Modified Soil-Adjusted Vegetation Index (MSAVI2) reduce soil-brightness effects and would be appropriate sensitivity checks for a follow-up assessment.

4.3 Land-use Confounders

A working sand dam can enable increased dry-season cultivation in its vicinity. Freshly tilled cropland reads as bare soil from satellite, and so an increase in dry-season farming would tend to reduce dry-season NDVI, even though the underlying intervention is succeeding. Without ground-truth on cultivation intensity and patterns, this analysis cannot fully disambiguate dam-induced cropping from dam failure.

4.4 No Ground-truth

The interpretation of the channel-aligned NDVI reduction as sand accumulation (§3.2) is based on the established physical mechanism of sand dams, not on direct field observation. A site visit with channel cross-section measurement and photographs of vegetation along the streambed would provide direct verification.

4.5 Statistical Power on Trends

Annual median dry-season vegetation fractions were computed for the treatment AOI and tested for trend using the Mann-Kendall test and the Theil-Sen slope estimator, both standard non-parametric methods for short noisy time series.⁹ With ten annual data points, no statistically significant trend is detected at the 5 per cent level, irrespective of the underlying truth. This is a sample-size limitation, not evidence that no trend exists.

5. What Would Be Required to Run a Treatment-Control Study

A defensible quantitative claim about the dam's effect on vegetation requires one or more carefully selected control areas. This section sets out the procedure that would be used in a follow-up assessment. The approach is standard in development-economics impact evaluation and in remote-sensing impact assessment.^{10,11}

5.1 Why a Properly Selected Control Is Required

A difference-in-differences (DiD) estimator subtracts a control area's pre/post change from the treatment area's pre/post change. The control absorbs shared regional drivers including climate, drought, and broader land-use trends. Validity rests on the assumption that the treatment and control areas would have evolved in parallel absent the intervention. This assumption is not given by inspection; it is testable empirically by comparing pre-construction trajectories. A control area that fails this test should not be used.

5.2 Tier 1 - Hard Exclusions

Any candidate control must clear these gates before being considered:

1. **Not on the same drainage as the treatment.** Verified using a digital elevation model (Copernicus DEM 30 m, free).¹²
2. **Outside the influence zone of any other sand dam.** Requires the implementing organisation's full registry of dam locations. All known dams should be buffered by 1 to 2 km and excluded.
3. **Same broad land cover.** Verified using ESA WorldCover 10 m or Google Dynamic World 10 m.^{13,14} Towns, plantations, irrigated fields, and water bodies are excluded.
4. **Same agro-climatic zone.** Verified using CHIRPS rainfall climatology⁸ – same average annual rainfall (within ± 10 per cent) and same bimodal pattern.
5. **Within 5 to 20 km of the treatment.** Close enough to share weather; far enough to be hydrologically independent.

5.3 Tier 2 - Soft Matching

Among Tier 1 survivors, candidates are matched on:

- Elevation band (within ± 100 m of treatment).
- Slope (within $\pm 2^\circ$).
- Aspect (compass orientation of the slope).
- Major soil class (FAO Harmonized World Soil Database, or ISRIC SoilGrids).¹⁵
- Pre-construction NDVI mean and seasonality (computed from Sentinel-2 2016 to October 2018).

5.4 Tier 3 - Pre-trend Validation

For each Tier 2 candidate, the pre-construction Sentinel-2 NDVI time series (2016 to October 2018) is compared to the treatment AOI's pre-construction series. Only candidates with a Pearson correlation above approximately 0.7 to 0.8 should be retained. The DiD assumption of parallel pre-trends is verifiable, not given. A candidate whose pre-trend does not track the treatment is not a valid control regardless of how similar it appears on paper.

5.5 Tier 4 - Multiple Controls

A single control is fragile to local quirks. The recommended procedure is:

- Select three to five candidates that pass Tiers 1 to 3.
- Run the full DiD with each control independently.
- Report the median DiD with the range across controls (for example, 'DiD = +6 ha, range across four controls: +2 to +9 ha').

A formal extension is the synthetic control method (Abadie 2003, 2010),¹⁰ which constructs a weighted combination of multiple candidates that best reproduces the treatment's pre-construction trajectory. The synthetic AOI is then used as the comparison.

5.6 What Would Be Required for the Machakos Dam

To run a defensible treatment-control study at this site:

- The implementing organisation's dam registry is the most valuable single input. Without it, no candidate control can be defensibly verified as outside another dam's influence zone.
- A small Python helper that automates Tier 1 gate checks (Copernicus DEM drainage, ESA WorldCover, CHIRPS climatology).
- Pre-trend validation across three to five candidates (already feasible from the existing Sentinel-2 pipeline).
- A re-run of the corridor analysis for each control with a hand-traced channel-like feature, so that the comparison becomes a true ring-by-ring DiD.

6. Conclusions

Across the analysis period, three patterns in the satellite record point to the dam having an effect on the local landscape.

The first is that loss of dry-season plant cover is concentrated on the river itself and falls off smoothly as you move away from it (§3.3). The riverbed lost about 28 percentage points of its dry-season green cover; land 500 to 1,000 metres from the river lost about 11. This kind of pattern, anchored to the river, is what would be expected from a working sand dam, and it would not be expected from drought alone, which would affect the whole area more uniformly.

The second is that the upstream and downstream halves of the river behave differently (§3.4). On the upstream side of the dam wall (the west-southwest arm, where water flows down towards the dam), the band of land 100 to 250 metres from the river lost only 14 percentage points of its dry-season green cover, the smallest reduction in any zone on that side; this is the riparian-benefit signature that would be expected from a working sand dam. On the downstream side (the north-northeast arm), the riverbed itself lost the most plant cover anywhere in the area (35 percentage

points), with the loss easing off smoothly with distance. The downstream channel reduction is larger than would be expected from the dam's mechanism alone (the dam wall does not pile sand on the downstream side) and is more easily explained by the channel-side vegetation having had the most to lose during the drought, or by dry-season cropping expanding along the downstream channel where the dam supports shallow water.

The third is that the analysis period was dominated by the worst regional drought in over 40 years, with annual rainfall at the site falling by 29 per cent. Sand dams are designed to deliver value during droughts, and the patterns in §3.3 and §3.4 describe how the dam's location appears to have shaped the local response to that drought.

These three patterns are evidence that the dam has had an effect on the surrounding land. They do not, on their own, say *how much* vegetation in absolute terms the dam preserved. The satellite imagery alone cannot rule out other explanations that would produce similar patterns, such as the drought hitting greener land harder, or the dam enabling more dry-season farming near the river. Distinguishing between these would require a site visit (channel sand depth measurement, photographs of riparian vegetation, conversations with farmers about cropping practices) and a parallel analysis of one or more comparable areas without dams. Section §5 sets out what the latter would involve.

7. Recommended Next Steps

To strengthen the assessment, the following steps are recommended in approximate order of how much each would add for the work required.

1. **Obtain the implementing organisation's register of dam locations.** This is the single most useful input the assessment is missing. Without it, no comparison area can be confirmed to be outside the influence of another dam.
2. **Run the multi-control study described in §5.** Once the dam register is available, three to five matched comparison areas can be selected and analysed using the existing pipeline. This is what would produce a defensible figure for how much vegetation the dam has preserved.
3. **Visit the dam site.** Direct observations of how much sand has built up behind the wall, photographs of plants growing along the riverbanks, and conversations with local farmers about dry-season cropping would distinguish between the three explanations for the patterns described in §3.2.
4. **Cross-check the satellite measurement.** The vegetation index used in this analysis (NDVI) is known to be sensitive to soil colour and moisture in addition to plant cover. Repeating the analysis with a soil-adjusted index (SAVI or MSAVI2) would confirm whether the patterns observed are real or partly an artefact of the index choice.
5. **Extend the analysis to other dams in the project.** If the Machakos dam is one of several built by the same project on the same drainage, repeating this analysis at each location and pooling the results would give a stronger overall picture than any single site can.

8. References

1. World Meteorological Organization. *State of the Climate in Africa 2022* (WMO-No. 1330). Geneva: WMO, 2023.
2. Ryan, C. and Elsner, P. 'The potential for sand dams to increase the adaptive capacity of East African drylands to climate change'. *Regional Environmental Change* 16, no. 7 (2016): 2087–2096. DOI: 10.1007/s10113-016-0938-y.
3. National Drought Management Authority (Kenya). *Drought Early Warning Bulletins for Machakos County, 2021–2023*. Nairobi: NDMA, 2023.
4. Microsoft. *Microsoft Planetary Computer: Sentinel-2 L2A collection*. <https://planetarycomputer.microsoft.com/dataset/sentinel-2-l2a>.
5. European Space Agency. *Sentinel-2 User Handbook*. ESA, 2015.
6. Huete, A.R. 'A soil-adjusted vegetation index (SAVI)'. *Remote Sensing of Environment* 25 (1988): 295–309. DOI: 10.1016/0034-4257(88)90106-X.

7. Hersbach, H. et al. 'The ERA5 global reanalysis'. *Quarterly Journal of the Royal Meteorological Society* 146, no. 730 (2020): 1999–2049. DOI: 10.1002/qj.3803.
8. Funk, C. et al. 'The climate hazards infrared precipitation with stations – a new environmental record for monitoring extremes'. *Scientific Data* 2 (2015): 150066. DOI: 10.1038/sdata.2015.66.
9. Hirsch, R.M., Slack, J.R. and Smith, R.A. 'Techniques of trend analysis for monthly water quality data'. *Water Resources Research* 18, no. 1 (1982): 107–121. DOI: 10.1029/WR018i001p00107.
10. Abadie, A. 'Using synthetic controls: feasibility, data requirements, and methodological aspects'. *Journal of Economic Literature* 59, no. 2 (2021): 391–425. DOI: 10.1257/jel.20191450.
11. Frontiers in Climate. 'Vegetation monitoring optimisation with NDVI and evapotranspiration using remote sensing measurements and land surface models over East Africa'. *Frontiers in Climate* 3 (2021). DOI: 10.3389/fclim.2021.589981.
12. European Space Agency. *Copernicus Digital Elevation Model*. <https://spacedata.copernicus.eu/collectio-digital-elevation-model>.
13. Zanaga, D. et al. *ESA WorldCover 10 m 2021 v200*. Zenodo, 2022. DOI: 10.5281/zenodo.7254221.
14. Brown, C.F. et al. 'Dynamic World, near real-time global 10 m land use land cover mapping'. *Scientific Data* 9 (2022): 251. DOI: 10.1038/s41597-022-01307-4.
15. Hengl, T. et al. 'SoilGrids250m: global gridded soil information based on machine learning'. *PLoS ONE* 12, no. 2 (2017): e0169748. DOI: 10.1371/journal.pone.0169748.

9. Technical Appendix

9.1 Methodology

9.1.1 Sentinel-2 Data, Cloud Filtering and Season Restriction Sentinel-2 L2A surface reflectance imagery was accessed via the Microsoft Planetary Computer.⁴ The full archive 2016-01-01 to 2025-12-31 was searched for the treatment AOI, year by year, returning approximately 580 candidate scenes.

Each scene's pixels were classified using the Sentinel-2 Scene Classification Layer. Pixels classified as cloud, cloud shadow, cirrus, snow, saturated, or no-data were masked. Scenes were retained for analysis only if at least 85 per cent of the AOI's pixels were classified as clear. After cloud filtering, 200 scenes were retained.

The corridor and per-pixel analyses were further restricted to scenes captured in the **long dry season** (June, July, August, and September of each year). After this restriction, 56 scenes remained. The long dry season is the period in which the dam's subsurface water storage is most diagnostic, since surface water is absent and any vegetation that holds its cover does so by drawing on stored or sub-surface moisture.² Pixel resolution is 10 m.

9.1.2 NDVI and the Vegetation Threshold The Normalized Difference Vegetation Index (NDVI), computed as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$ from Sentinel-2 bands B08 and B04, is a widely used measure of green vegetation cover from satellite imagery.⁵ NDVI ranges from approximately -0.2 (water, bare soil) to +0.9 (dense vegetation).

For each retained scene, every 10 m × 10 m pixel was classified as vegetated if NDVI exceeded 0.30. This threshold is a conventional cutoff for moderate vegetation cover and is in line with the Sentinel-2 vegetation literature.^{5,6} NDVI is known to be sensitive to soil brightness and to saturate in dense vegetation; both limitations are relevant to a semi-arid site such as Machakos and are discussed in §4.

9.1.3 Pre-construction vs Post-construction Comparison For the treatment AOI, scenes were partitioned by date relative to the dam construction:

- **Pre-construction (control period):** scenes captured before 1 October 2018.
- **Post-construction (treatment period):** scenes captured after 1 October 2018.

The October 2018 construction month was excluded from both periods.

For each pixel, in 10-day day-of-year windows, the analysis computed the fraction of clear scenes in each period in which the pixel exceeded the NDVI threshold. The post-construction fraction minus the pre-construction fraction gives the change in vegetation frequency for that pixel and that day-of-year window. Averaging across the 37 day-of-year bins (and ignoring bins with no data on either side) gives a single per-pixel post-pre change. This pixel-level change is used in §3.2.

The day-of-year binning controls for seasonality: like-for-like seasonal positions are compared across years rather than mixing wet-season and dry-season scenes.

This ‘pre-construction as control, post-construction as treatment’ framing is a temporal contrast within a single area. It is not a treatment-control study in the spatial sense (one area with the intervention, one without). What that would require is set out in §5.

9.1.4 Channel Trace and Distance Bands The streambed visible in the AOI was traced by manual interpretation of dry-season Sentinel-2 imagery in Google Earth. The trace was produced as two KML files: `sanddam.kml` covers the west-southwest arm of the river (upstream of the dam wall, see flow-direction note below), and `sandam-upstream.kml` covers the north-northeast arm (downstream of the dam wall). The two files are combined into a single multi-line geometry for the analysis. The combined polyline contains 38 waypoints over approximately 1,673 m.

The combined geometry was reprojected from WGS84 to UTM 37S so that distances are in metres. Concentric ring buffers were computed at five distance bands: 0–50 m, 50–100 m, 100–250 m, 250–500 m, and 500–1,000 m from the combined channel geometry. Each ring was rasterised onto the same pixel grid as the cached NDVI arrays. Figure 1.1 (right panel) shows the rings.

For each ring, the analysis computed the mean per-pixel post-pre change across all pixels in the ring. The resulting five values give a distance-decay profile for the dam site.

A working sand dam is expected to produce a spatially structured pattern around the channel, since the hydrological mechanism (raised water table) is strongest near the streambed and weakens with distance. Random measurement noise and uniform regional climate effects are not expected to produce a distance-structured pattern.²

Flow direction. The flow direction along the channel was confirmed by querying the SRTM 30 m digital elevation model¹² at sampled waypoints along each arm of the channel. The west-southwest arm of the river rises 9 metres from the dam to its far end (statistically meaningful given SRTM ~5 m vertical accuracy) and is therefore upstream of the wall. The north-northeast arm is essentially flat from the dam (1 metre difference, within DEM noise) and is downstream of the wall. The two KML file names were assigned before the elevation check and do not match the actual flow direction; the file names are kept as the user wrote them, and the analysis code maps each file to the correct upstream / downstream role.

9.1.5 Precipitation Context Daily precipitation totals at the dam coordinates were obtained from the Open-Meteo Historical Weather API, which serves ERA5 reanalysis data.⁷ Daily values were aggregated to monthly totals for the time-series figure (§3.1). For follow-up work, CHIRPS8 would be the preferred source because it merges satellite estimates with rain-gauge data tuned for African drought monitoring.

9.2 Code, Data, and Outputs

- **Analysis package:** `backend/exploration/`
 - `config.py` (centralised configuration)
 - `fetch.py` (STAC search and per-AOI cache build)
 - `cache.py` (zarr open and atomic writes)
 - `indices.py` (NDVI, MNDWI, valid-pixel mask)
 - `aggregate.py` (per-pixel pre/post fractions, day-of-year binning, ring buffers)
 - `stats.py` (Mann-Kendall, Theil-Sen, Mann-Whitney, DiD scalars)
 - `precipitation.py` (Open-Meteo / ERA5 daily precipitation)
 - `visualisation.py` (all rendering)
- **Driver scripts:** `backend/exploration/scripts/`
 - `step01_build_cache.py` (slow, network – populates zarr cache)

- step02_fetch_precipitation.py (network – daily precipitation cache)
- step03_run_analysis.py (fast – runs all numerical analysis from cache)
- step04_render_outputs.py (fast – produces all figures)
- **Cache:** backend/_cache_fulltimeseries/ (approximately 120 MB, version-keyed)
- **Outputs:** backend/exploration/outputs/ (all figures, including corridor_decay_by_segment.png, plus CSV / JSON summaries including corridor_by_segment.csv)
- **Reproducibility:** with cache populated, the full re-run takes under one minute. Cache invalidates automatically if the cloud filter, band list, or DOY bin size changes.

9.3 Configuration

Parameter	Value
Vegetation threshold	NDVI > 0.30
Cloud filter	≥ 85 per cent of AOI pixels classified as clear
Season filter	June, July, August, September (long dry season)
Construction date (pre/post split)	1 October 2018
Drought window	1 October 2020 to 30 June 2023
Day-of-year bin size	10 days (37 bins per year)
Pixel resolution	10 m × 10 m
Treatment AOI scenes after cloud filter	200
Treatment AOI scenes after season filter (Jun-Sep)	56
Channel polyline (combined)	hand-traced, two segments, 38 waypoints, approximately 1,673 m total (backend/sanddam.kml + backend/sandam-upstream.kml)

9.4 Methodological Notes

Within-AOI distance-decay (§3.3) is the central test of this assessment. It compares ring-mean changes within a single AOI to test for a spatial signature of an intervention. It is robust to uniform region-wide drivers, because uniform drivers do not produce distance-structured patterns.

Difference-in-differences (DiD) is a standard causal-inference approach in environmental impact evaluation. It is not used as a primary estimator in this assessment because no defensibly selected control area was available; §5 sets out the procedure for follow-up work.

Mann-Kendall and Theil-Sen are non-parametric trend tests appropriate for short, noisy time series with seasonal structure and outlier years.⁹

Document version 6.4 – §1 (Site and Approach) further simplified: the technical detail about how the channel was traced (KML files, waypoint counts, polyline length) has been moved into §9.1.4 in the technical appendix. The §1 body now refers a reader to that section for the full technical detail. The corrected file-to-segment orientation (sanddam.kml = upstream WSW, sandam-upstream.kml = downstream NNE) is applied consistently in the moved description.